Data Management Project

Enterprise Data Quality and Intelligence Platform with ML-Based Anomaly Detection

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TABLE OF CONTENT

Introduction……..3

Background……..3

Aims and Objectives……4

Plan of Implementation……6

Milestones and Timeline…...……...……8

Conclusion……..10

References……...11

INTRODUCTION

Organizations today rely more, than ever on data to run their operations generate reports and make important decisions. The data they use often has problems—missing values, duplicate entries, wrong formats, inconsistent details and strange or unusual values. These issues can make reports unreliable and lead to business choices.

To address this we are proposing an Enterprise Data Quality and Intelligence Platform that uses machine learning to detect anomalies. This platform will serve as a centralized tool to check and improve the quality of business datasets. Users will be able to upload their data examine its structure find quality issues apply rules to validate the data clean it up and see quality results in time through an interactive dashboard.

A key feature of the platform is its use of machine learning to find data records that standard validation rules might miss. These records will be flagged for review so users can decide if they need to be corrected. The platform will be tested using both controlled datasets to measure how well it detects anomalies how much it improves data quality and how fast it processes data.

In addition to the existing data-quality management features, this project will use multiple structured datasets from Sri Lankan sources to evaluate the proposed platform. Datasets from areas such as population, agriculture, employment and economic activities will be considered. These datasets will be used to investigate data-quality issues and develop and evaluate anomaly-detection models. The project will also compare different statistical and machine-learning algorithms to identify unusual patterns across different datasets.

BACKGROUND

Organizations gather data from places like transaction systems, spreadsheets, customer databases and inventory management tools. Keeping this data clean and accurate is hard. Records might be missing information appear than once have wrong formats or don’t match up across systems.

Old ways of cleaning data often rely on people checking each record by hand or using set rules. These rule-based checks work well for catching issues.. They can miss strange or unusual records that don’t break any specific rule. Also tasks like data profiling, cleaning, validation and reporting are usually done with tools. This makes it tough to keep track of improvements and stay consistent over time.

The new platform aims to bring all these steps in one place. It will mix data-management methods with machine learning that detects unusual patterns. This helps users find problems review records easily and check data quality in a clear and structured way. The platform will start with business data—, like customer records, sales data and inventory data.

AIMS AND OBJECTIVES

Primary Aim:
To build an enterprise data quality platform that brings together automated data profiling, validation, cleaning, quality assessment and machine-learning-based anomaly detection. This platform will help teams spot issues early keep data accurate and make decisions. It's designed to work across systems and handle large volumes of data smoothly. The goal is to make data trustworthy and reliable, for everyone using it.

Objectives:

1.Automated Data Profiling and Quality Assessment:
Build a data profiling module that automatically checks any dataset. It looks for missing values, duplicate records, wrong data types, format inconsistencies and odd statistical patterns. It then puts together a summary of data-quality problems so that you can see how your data looks.

2.User-Defined Data Validation:
Create an interface that lets you upload a dataset and set validation rules that fit your needs.You can set limits on fields check that email addresses follow a valid pattern specify acceptable number ranges and enforce unique identifiers.

3.Automated Data Cleaning and Transformation:
Add cleaning features that fill or remove values spot duplicate records make formats the same and fix data types. You stay in control of each cleaning step. The original dataset stays untouched.

4.Machine Learning-Based Anomaly Detection:
Use a machine-learning model like Isolation Forest to spot records that look odd compared to the rest of the data. The system will flag anomalies that go beyond your set rules and you can review each flagged record.
The project will initially implement the Isolation Forest algorithm and extend the anomaly-detection module by incorporating Local Outlier Factor (LOF) and One-Class SVM. Statistical techniques such as Z-score and Interquartile Range (IQR) will also be used as baseline approaches. These algorithms will be trained or configured using suitable features from selected UCI benchmark and Sri Lankan datasets. Their anomaly detection performance will be compared using appropriate evaluation metrics.

5.Data Quality Scoring and Visualization:
Build a scoring system that rates data quality on completeness, validity, consistency and uniqueness. Put the scores on a dashboard that shows detected problems and lets you compare the original data with the cleaned version.

6.Data Lineage and Processing History:
Set up a log that keeps track of where the data came from the results of each validation every cleaning step and the time each change was made. This history helps you trace every change and see how your data has evolved.

7.Reporting and Dataset Export:
Add a feature that creates reports that list all problems, quality scores and the cleaning outcomes. You can also export the cleaned data, as CSV or Excel files.

8.Scalability and Processing Efficiency:
Make sure the platform can handle large datasets without slowing down too much. Test it with sized data and look for ways to make it faster.

9.Advanced Data Quality Intelligence (Future Scope):
Plan to grow the platform by adding anomaly detection, better ways to fill missing values and tools that find duplicate records by comparing how similar the data looks.

Working Methodology:

Step 1: Upload Dataset

The user starts by uploading a dataset to the platform.Supported file formats include CSV and Excel. These are types most people use.The system checks the file to make sure it's okay. Gets it ready for analysis.

Step 2: Data Profiling and Analysis

The system looks at the dataset to figure out its structure. It checks column names, what kind of data's in each column how many records there are and how the data is spread out.It also finds problems like missing values, duplicate rows, wrong formats or values that don't match expected patterns.All this information is shown in a profiling summary. This helps users understand what they're working with before they make any changes.

Step 3: Data Validation and Issue Detection

The system runs checks based on rules that're either built-in or set by the user.These rules can catch missing required fields, invalid email addresses, numbers that are high or too low or duplicate IDs.The platform shows exactly which records failed the checks and explains what rule was broken.

Step 4: Machine Learning-Based Anomaly Detection

Use Scikit-learn to implement and experiment with multiple anomaly detection algorithms, including Isolation Forest, Local Outlier Factor (LOF), and One-Class SVM. Prepare suitable features and apply preprocessing steps such as scaling and encoding. Statistical techniques such as Z-score and Interquartile Range (IQR) will also be considered as baseline approaches. Evaluate and compare the anomaly detection results using suitable test data and performance metrics such as precision, recall, and F1-score where labelled data are available.

Step 5: Data Cleaning and Transformation

The user picks cleaning steps based on the issues found.The system does things like filling in missing values checking for duplicates making formats consistent and fixing data types.The original dataset is kept safe. This way users can compare the cleaned version with the original. Avoid losing data by accident.

Step 6: Data Quality Scoring and Visualization

The system calculates scores for how complete consistent and unique the data is.A dashboard shows these scores before and after cleaning.Charts and summaries help users see which problems were fixed and which ones still need attention.

Step 7: Report Generation and Export

The platform creates a report with data-quality scores, validation errors, anomalies. A summary of cleaning steps.Users can download both the cleaned dataset and the report.The system also saves a record of what was done so users can refer back, to it later.

Example Workflow:

1. The customer dataset uploaded by the business analyst consists of 5,000 records.

2. The system detects problems like absent emails, duplicated customer IDs, invalid email addresses, and strange age values.

3.The user checks the issues found and runs another module based on machine learning for further discovery of unusual records.

4. After selecting the required cleaning method, the platform conducts data preparation but ensures the data is kept unchanged.

5. The dashboard shows the contrast between the quality of the dataset before and after the cleaning process.

6. The cleaned data along with the report that covers issues found, cleaning methods used and improvements achieved is available for download.

This scenario highlights how the proposed application blends traditional data management technologies with machine learning capabilities.

IMPLEMENTATION PLAN

1. Research and Dataset Preparation

Start by learning about data-quality management. Look into data profiling, data validation, data cleaning and how to measure data quality. Find datasets that reflect real business situations like customer details sales records and inventory data. Create test datasets with known data-quality issues. These will help test how well the system detects problems.

The project will prioritise publicly available Sri Lankan datasets obtained from official sources, including the Sri Lanka Open Data Portal, Department of Census and Statistics, Department of Agriculture and Central Bank of Sri Lanka. Datasets representing different domains, such as population, agriculture, employment and economic activities, will be investigated.

Suitable datasets will be selected based on their availability, data format, number of records, attributes and suitability for anomaly detection. Each dataset will be separately preprocessed and used for model development and evaluation. Controlled test datasets containing known data-quality issues will also be prepared to assess the performance of the proposed algorithms. In addition to local datasets, internationally recognized benchmark datasets, including the UCI Adult and Bank Marketing datasets, will be used for model experimentation and comparative evaluation.

2. Data Profiling and Validation Module

Use Python and Pandas to examine dataset structures. Check data types missing values, duplicate entries and statistical patterns. Build validation rules to spot data-quality problems. Make it possible to set up and apply these rules to datasets as needed.

3. Data Cleaning and Quality Scoring

Write functions in Python and Pandas to. Transform data. Create scoring systems to rate data on completeness, validity, consistency and uniqueness. Compare quality scores before and after cleaning to see how much improvement happens.

4. Machine Learning Model Development

Use Scikit-learn to implement and experiment with multiple anomaly detection algorithms, including Isolation Forest, Local Outlier Factor (LOF), and One-Class SVM. Prepare suitable features and apply preprocessing steps such as scaling and encoding. Statistical techniques such as Z-score and Interquartile Range (IQR) will also be considered as baseline approaches. Evaluate and compare the anomaly detection results using suitable test data and performance metrics such as precision, recall, and F1-score where labelled data are available.

5. System Development

Frontend: Build a web interface with React. It should allow users to upload datasets set validation rules, view dashboards and see results.

Backend: Use Python with FastAPI to handle business logic process data and link system parts.

Database: Use PostgreSQL to store metadata, validation rules, quality reports and history of operations.

Visualization: Use Plotly to show charts and compare data-quality metrics.

6. Testing and Optimization

Test the system with datasets and datasets, with known errors. Check how accurate validation is, how well anomalies are detected, how much data quality improves and how long processing takes. Use datasets of sizes and optimize speed and efficiency where needed.

The anomaly-detection algorithms will be evaluated and compared using multiple selected datasets. Precision, recall and F1-score will be calculated using controlled test data with known anomaly labels. Processing time and memory requirements will also be measured to investigate the efficiency of the algorithms. The results will help identify the strengths and limitations of each approach across different dataset types.

7. Documentation and Demonstration

Write system requirements, design architecture diagrams, database models and implementation notes. Explain the machine-learning method, evaluation process and results. Show the system using real business data examples.

- The tools and technologies used can be changed based on project needs, available resources or how the project progresses.

MILESTONES AND TIMELINE

This project consists with 2 phases

I. Semester 6 phase
II. Semester 7 phase

Table 1: Project Timeline Phase 1

Week	Milestone	Duration	Activities
Semester 6: Weeks 1–4	Research and Planning	4 weeks	Study data quality management, data profiling, data validation, data cleaning, and machine learning-based anomaly detection. Review related research, investigate anomaly detection algorithms, and define the project requirements.
Semester 6: Weeks 5–8	Dataset Preparation and System Design	4 weeks	Identify and collect suitable UCI benchmark datasets, including Adult and Bank Marketing, together with suitable Sri Lankan datasets. Prepare controlled datasets containing known data-quality

			issues, examine dataset characteristics, and design the system architecture and database schema.
Semester 6: Weeks 9–12	Data Profiling, Validation and Initial ML Development	4 weeks	Develop the dataset upload module, implement automated data profiling, and create validation rules to identify missing values, duplicates, invalid formats, and inconsistent records. Begin initial anomaly detection experiments using Isolation Forest and investigate other suitable algorithms. Start frontend and backend integration.
Semester 6: Weeks 13–14	Initial Testing and Review	2 weeks	Test the initial data-processing modules and anomaly detection experiments. Investigate Local Outlier Factor (LOF) and One-Class SVM, review the implementation against the requirements, and integrate the initial machine learning and data-processing components.

Table 2: Project Timeline Phase 2

Week	**Milestone**	**Duration**	**Activities**
Semester 7: Weeks 1–4	Data Cleaning and Quality Scoring	4 weeks	Implement data-cleaning and transformation functions, develop data-quality scoring methods, and compare quality measurements before and after cleaning. Continue integrating the data-processing modules with the application.
Semester 7: Weeks 5–7	Advanced Machine Learning	3 weeks	Further develop and integrate Isolation Forest, Local Outlier Factor (LOF), and One-Class SVM. Implement statistical baseline techniques such as Z-

	Development and Comparison		score and Interquartile Range (IQR). Compare anomaly detection results using the selected benchmark and Sri Lankan datasets.
Semester 7: Weeks 8–10	System and Dashboard Development	3 weeks	Complete the integration of the React frontend, FastAPI backend, and PostgreSQL database with the data-processing and machine learning modules. Develop interactive dashboards, processing history, and data-quality report generation.
Semester 7: Weeks 11–12	Testing and Validation	2 weeks	Test the integrated system using selected UCI benchmark, Sri Lankan, and controlled datasets. Evaluate validation results, anomaly detection performance, data-quality improvements, and processing time using suitable evaluation measures.
Semester 7: Weeks 13–14	Optimization and Final Review	2 weeks	Optimize system performance, resolve identified issues, complete project documentation, and prepare the final system demonstration and presentation.

CONCLUSION

The proposed Enterprise Data Quality and Intelligence Platform is designed to offer a solution for making sure that structured business datasets are accurate and trustworthy. The system will bring together data profiling, rules-based checks, data cleaning, data quality ratings and machine learning methods for finding strange data patterns in one single place.

The platform will be tested using a mix of well-known international benchmark datasets and local datasets from Sri Lanka to evaluate its performance across different data domains.A range of methods for finding data like Isolation Forest, Local Outlier Factor (LOF) and One-Class SVM will be studied and compared with traditional statistical techniques to find unusual patterns in data.

With the help of dashboards detailed data quality reports and a record of past processing steps the platform will allow users to see what problems exist with their data and make better choices, about how to fix them. The main goal of this project is to create a system that can grow and be used in situations, which helps with reliable data analysis and better decisions based on data.

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